



Security Visibility Tradeoffs in TLS-Encrypted Cloud Network Environments

Presented by Michael Fisher
M.S. Cyber Security
Graduate Project
Spring 2026

Background and Problem

TLS Encryption

- Protects **confidentiality** of data in transit
- Encrypts application-layer payloads
- Widely adopted in modern web traffic (HTTPS)

Security Challenge

- Encrypted traffic **hides packet contents**
- Signature-based detection becomes limited
- IDS cannot inspect request/response data

Impact on Detection

- **Reduced visibility** into application-layer attacks
- Detection shifts to metadata and behavior
- Potential gap in threat identification

Research Question and Hypothesis

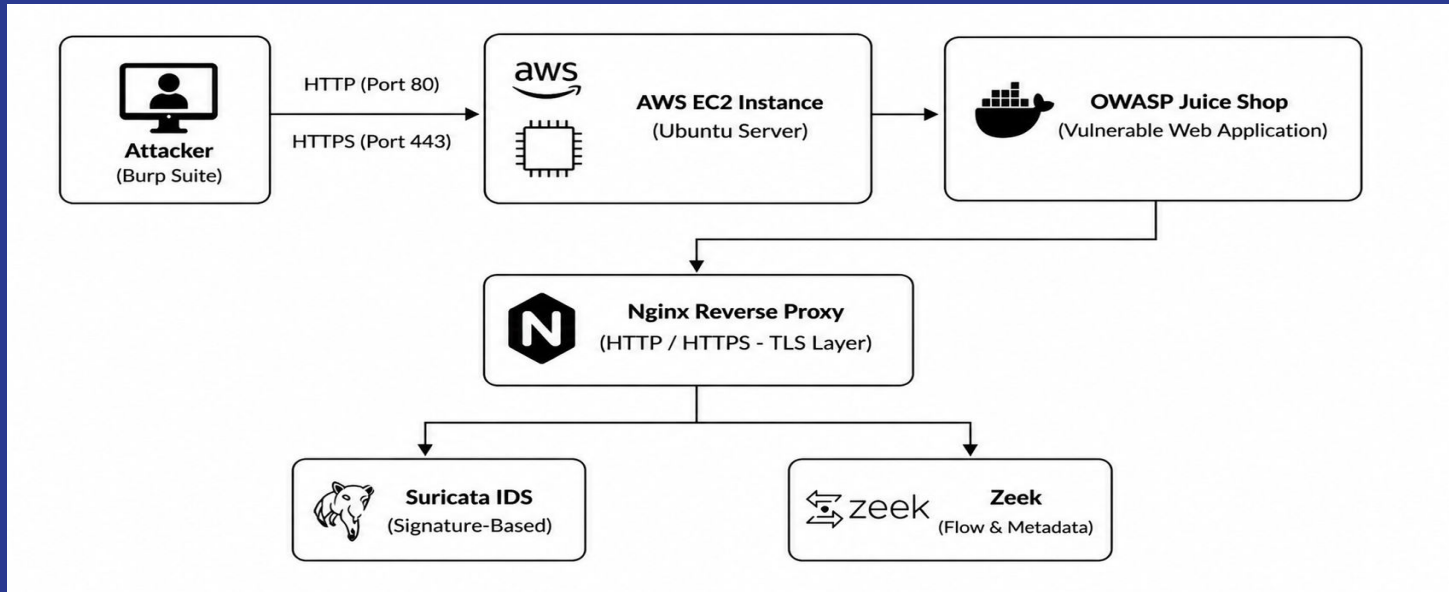
Research Question

How does TLS encryption impact the visibility and detection capabilities of network-based intrusion detection systems in cloud environments?

Hypothesis

TLS encryption reduces IDS effectiveness for application-layer attacks by removing payload visibility, while preserving limited metadata and flow visibility.

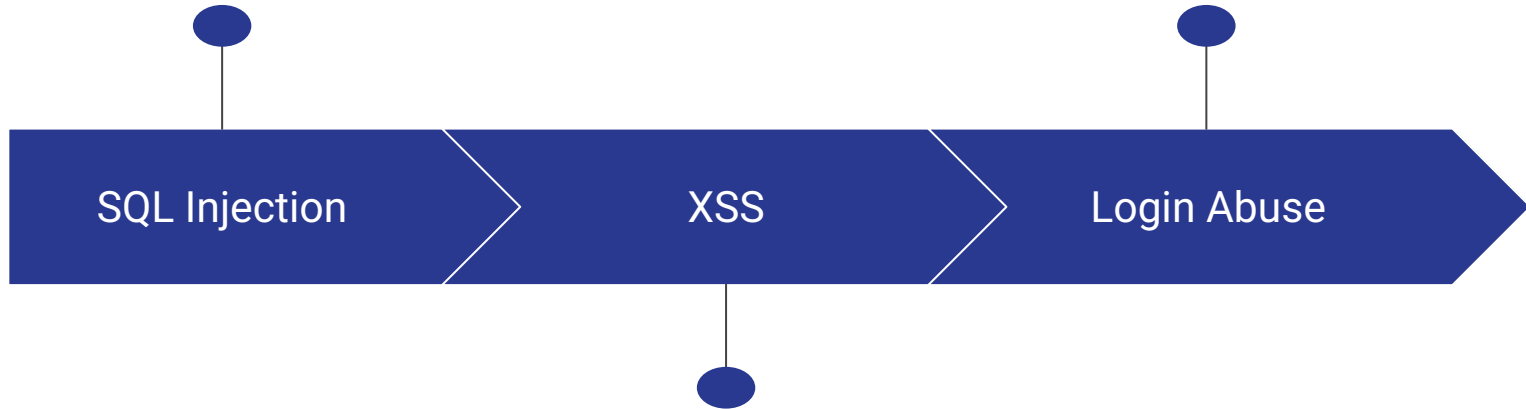
Lab Setup



- AWS EC2-hosted OWASP Juice Shop target
- Suricata and Zeek used for monitoring
- Burp Suite used to generate identical HTTP and HTTPS attacks

Modified login request through
Burp

Repeated failed login attempts



Injected script into search
parameter

ATTACK TYPES

Method Control : Same target, same attacks,
only TLS changed

TTT Results

- Payload visible in transit
- Suricata able to inspect request content
- Zeek recorded application-layer and connection details
- HTTP provided strong visibility for traffic analysis

The image displays a network analysis environment. On the left, a terminal window shows Suricata logs for a host at 172.31.41.99. The logs indicate that Suricata is processing HTTP requests and detecting unauthorized access attempts. On the right, a Wireshark packet capture shows an HTTP GET request to /api/users/whomami. The packet details pane shows the request structure, including the method, URL, and headers. The packet bytes pane shows the raw data of the request, including the payload.

Terminal window (left) shows Suricata logs for a host at 172.31.41.99. The logs indicate that Suricata is processing HTTP requests and detecting unauthorized access attempts.

Wireshark packet capture (right) shows an HTTP GET request to /api/users/whomami. The packet details pane shows the request structure, including the method, URL, and headers. The packet bytes pane shows the raw data of the request, including the payload.

HTTPS Results

- Payload encrypted and no longer visible
- Suricata lost application-layer inspection capability
- Zeek retained TLS and connection metadata
- Detection shifted from content to behavior

The image displays two side-by-side screenshots from network security tools. The left screenshot shows the Burp Suite interface with the 'HTTP history' tab selected, displaying a list of intercepted requests. The right screenshot shows the Wireshark interface with the 'Packet Details' pane expanded, showing the 'HTTP' section of a captured packet. The Wireshark packet list shows a 'GET /api/quantify/' request from 172.31.41.99 to 172.31.41.99. The packet details pane shows the 'Request' section with a 'GET /api/quantify/' method and a 'Content-Type: application/json' header. The 'Response' section shows a '200 OK' status and a 'Content-Type: application/json' header. The 'Raw' section shows the raw packet data in hexadecimal and ASCII.

Comparison and Conclusion

TLS significantly reduces IDS detection visibility at the application layer while preserving limited metadata and behavioral insight.

HTTP vs HTTPS Visibility Comparison

Feature	HTTP	HTTPS
Payload Visibility	Yes	No
Suricata content inspection	Yes	Limited
Zeek metadata visibility	Yes	Yes
Application-layer attack visibility	High	Reduced

References

- [1] M. Qutqut et al., “A comparative evaluation of Snort and Suricata for detecting data exfiltration tunnels in cloud environments,” *Journal of Cloud Computing*, 2024.
- [2] R. Zheng, J. Liu, L. Liu, S. Liao, K. Li, J. Wei, L. Li, and Z. Tian, “Two-layer detection framework with a high accuracy and efficiency for a malware family over the TLS protocol,” *PLOS ONE*, vol. 15, no. 5, 2020.
- [3] B. Anderson, “Detecting Encrypted Malware Traffic (Without Decryption),” Cisco Security Blog.
- [4] R. Antonello et al., “Deep packet inspection tools and techniques in commodity platforms: Challenges and trends,” *Journal of Network and Computer Applications*, 2012.
- [5] P. Zhang et al., “Detection and utilization of new-type encrypted network traffic in distributed scenarios,” *Engineering Applications of Artificial Intelligence*, 2023.

